

Chapter 1

The nature of econometrics and economic data

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Outline

- **What is econometrics**
- **What can econometrics do for us**
- **Data: The input**
- **Steps in Empirical Analysis**

Econometrics in our daily life

- 好男人都结婚了吗？ ---探究我国男性工资婚姻溢价的形成机制

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- 在控制相关特征变量之后,在婚男性工资比非在婚男性工资显著高出6.8%。(婚姻溢价)

Econometrics in our daily life

- 为什么存在溢价？
 - 分工理论：结婚→分工合理→工资高
 - 相夫(选择)理论：条件好→工资高→结婚
- 分工理论：婚姻没有通过减少男性家务时间提高其工资
- 相夫效应：妻子的受教育程度和妻子收入水平显著正向影响丈夫工资

What is econometrics

- Ragnar Frisch (1933): “The (econometric) society... main object shall be to promote studies that aim at a unification of the **theoretical**-quantitative and **empirical**-quantitative approach to economic problems that are penetrated by constructive and rigorous thinking similar to that which has come to dominate in the natural sciences.”

What is econometrics

- Econometrics is not just statistics, nor economic theory, nor application of mathematics to economics. It is the unification of the three.
- Econometrics focuses on the problems inherent in collecting and analyzing **non-experimental** economic data.

What can econometrics do for us?

- Testing economic theories
- Explain behaviors that are of economic nature
- Explain behaviors disguised not to be of economic nature
- Policy evaluation

What can econometrics do for us?

- Example: criminal behavior

$$y = f(x_1, x_2, x_3, x_4, x_5, x_6, x_7)$$

y : hours spent in criminal activity

x_1 : "wage" of the criminal activity

x_2 : hourly wage in legal employment

x_3 : Income other than from crime or employment

x_4 : Probability of getting caught

x_5 : Expected sentence if got caught

x_6 : Age

x_7 : Discrimination degree received in cities

What can econometrics do for us?

- In-class question:

What are the factors of house prices?

- ✓ Mentimeter

<https://www.menti.com>

Type in code and vote

What can econometrics do for us?

- Overall, we use econometrics to explain phenomena of economic nature, make policy recommendations and make forecasts about the future.

Data : The Input

1. Cross – Section data

- A cross-sectional data set consists of a sample of individuals at **a given point in time**.

$$\{Y_1, Y_2, \dots, Y_N\}, i = 1, 2, \dots, N$$

- Widely used in economics, especially in fields like labor economics, health economics, urban economics, etc.
- Cross-sectional data can include variables that describes situations in the past or in the future.

Table 1 A Survey of Rural Households' Income in Province S at Year 2004

户主姓名	2004 年家庭纯收入 (元)	2004 年家庭现金纯收入 (元)	户主姓名	2004 年家庭纯收入 (元)	2004 年家庭现金纯收入 (元)
田	12000	12000	李	10000	10000
杨	11000	11000	杜	10000	10000
何	12000	10000	吴	15000	15000
郭	11000	11000	王	17000	17000
薛	8000	8000	何	15000	14000
李	10000	7000	王	20000	20000
杨	9000	8000	严	20000	15000
芑	8000	8000	张	30000	30000
马	10000	10000	姜	25000	25000
史	8000	6000	张	38000	15000
何	7000	7000	张	4000	4000
张	10000	8000	张	2000	2000

Data : The Input

- Cross-sectional data structure often makes **random sampling** assumption reasonable.
- *Random sampling*: When $\{Y_1, Y_2, \dots, Y_N\}$ is a random sample from the density $f(y)$, Y_i are *independent, identically distributed (i.i.d.)* random variables from $f(y)$.

Data : The Input

2. Time – Series data

- A Time – Series data consists of observations on **a variable** or several variables **over time**.

$$\{Y_1, Y_2, \dots, Y_T\}, t = 1, 2, \dots, T$$

Table 2 China's GDP, Currency and M1, 1978 – 2002

Year	GDP	M1(亿元)	Year	GDP	M1(亿元)
1978	100	954.7	1990	283	8133.3
1979	107.6	1208.1	1991	308.8	10061.5
1980	116	1486.2	1992	352.2	14294.5
1981	122	1707	1993	398.4	18167.5
1982	133.3	1972.9	1994	448.7	20540.7
1983	148.2	2291.5	1995	489.1	23987.1
1984	170.9	3233	1996	536.8	28514.8
1985	193.5	3450.1	1997	582.9	34826.3
1986	209.9	4393.8	1998	628.4	38953.7
1987	234.1	5173.1	1999	673.5	45837.3
1988	260.5	6376.4	2000	730	53147.2
1989	271.5	6804.6	2001	782.6	59871.6
1990	283	8133.3	2002	848.9	70881.8

Data : The Input

- Time Series data has three important differences from cross-sectional data.
 - The ordering of observations in Time series data is meaningful.
 - Observations are often correlated.
 - Data frequency needs special attention.

Data : The Input

3. Pooled cross sections

- Pool both cross-sectional and time series data.
- Pooling cross sections from different years is often an effective way of analyzing the effects of a new government policy.

Eg: two cross-sectional household surveys are taken.

Table 3 Pooled cross sections: two years of housing prices

Obsno	Year	Hprice	Proptax	sqrft	bdrms	bthrms
1	1993	88500	42	1600	3	2
2	1993	67300	36	1440	3	2.5
3	1993	134000	38	2000	4	2.5
.						
.						
250	1993	243600	41	2600	4	3
251	1995	65000	16	1250	2	1
252	1995	182400	20	2200	4	2
253	1995	97500	15	1540	3	2
.						
.						
520	1995	57200	16	1100	2	1.5

Data : The Input

4. Panel Data

- Panel data is a sets of time-series observations for a group of individuals.
- Her other English names: longitudinal data, time-series cross-section data.
- Her Chinese names: 面板数据, 板块数据, 综列数据。

Table 4 Some Variables of Welfare Recipients in State A of country U

time	person	earnings	hour	employ	gender
19994	3	0	0	0	F
20001	3	0	0	0	F
20002	3	0	0	0	F
20003	3	0	0	0	F
19983	7	6284.25	471	1	F
19993	7	0	0	0	F
20001	7	0	0	0	F
20002	7	3425.42	280	1	F
19984	10	345.08	0	1	F
20004	10	2686.48	347	1	F
19982	18	226.6	44	1	F
19993	25	0	0	0	F
19994	25	289.25	45	1	F
20001	25	2226.76	259	1	F
19991	37	412.94	8	1	F

Data : The Input

- Because panel data require replication of the same units over time, panel data sets are more difficult to obtain than pooled cross sections.
- The advantage of panel Data
 - Having multiple observations on the same units allows us to control for certain unobserved characteristics of individuals, firms, and so on.
 - Allows us to study the importance of lags in behavior or the result of decision making.

Steps in Empirical Econometric Analysis

1. Formulate the question of interest carefully, so that it is specific enough to be able to be answered by the data at hand and through using econometric techniques available.
2. Collect needed data for the question of interest.

Steps in Empirical Econometric Analysis

3. Use basic economic tools to formulate an economic model that describes the relations between the main variables of interest.
4. Use knowledge about model specification to transform an economic model into an econometric model that is ready for estimation and prediction.

Steps in Empirical Econometric Analysis

- Make important assumptions before estimation of parameters of interest.
- The *ceteris paribus* Assumption
 - Assumes other things being equal during the process of the econometric analysis.
 - When carefully applied, econometric methods can simulate a *ceteris paribus* experiment.

Steps in Empirical Econometric Analysis

5. Specify hypothesis of interest in terms of the unknown parameters .
6. Use econometric methods to estimate the parameters and formally test the hypotheses of interest .

为什么存在男性工资婚姻溢价？

1. 分工理论 vs. 相夫(选择)理论
2. Pane data: 1989—2009年中国健康与营养调查数据(CHNS)，包括收入、受教育程度、配偶受教育程度、配偶收入水平等等
3. $M_{wage} = f(Z, X_1, X_2)$
 X_1 : 男性家务劳动时间
 X_2 : 配偶受教育程度、配偶收入水平
4. $\ln M_{wage} = \alpha + \beta Z + \gamma X_1 + \delta X_2 + \epsilon$

为什么存在男性工资婚姻溢价？

- 5. 分工理论： $\gamma \neq 0, \delta = 0$
 相夫(选择)理论： $\gamma = 0, \delta \neq 0$
- 6. 估计参数 γ, δ

Summary

1. What is econometrics?
2. Data structure
 - Cross-section
 - Time-series
 - Pooled
 - Panel
3. Steps in empirical analysis